

E056

Budgeting and Cost Control

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A city IT department is estimating a new fiber-network rollout. Reliable records show similar installs cost \$18 per linear foot, and the new network is 40,000 feet long. The department head drives a hybrid and prefers vendor-neutral designs. Which technique and estimate fit?

- A. Bottom-up estimating; \$18.
- B. Parametric estimating; \$2,222.
- C. Parametric estimating; \$720,000.
- D. Analogous estimating; \$720,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A project's work-package estimates total \$900,000. The team adds \$60,000 of contingency for identified risks, and leadership holds \$40,000 of management reserve for unknown risks. The sponsor is a former CFO who reviews every invoice. What is the cost baseline?

- A. \$940,000
- B. \$900,000
- C. \$960,000
- D. \$1,000,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A project budget is \$2.5 million, including \$200,000 of management reserve. Contingency reserve inside the baseline is \$150,000. What is the cost baseline?

- A. \$2.5 million
- B. \$2.3 million
- C. \$2.15 million
- D. \$2.7 million

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

At the halfway review, a warehouse-automation project has EV of \$300,000, AC of \$375,000, and PV of \$320,000. The project team just won an internal innovation award. What is the CPI, and what does it mean?

- A. 1.25; the project is under budget.
- B. 0.94; the project is slightly behind schedule.
- C. -\$75,000; the project is over budget.
- D. 0.80; the project gets 80 cents of work per dollar spent and is over budget.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A project's CPI is 0.90 and its earned value is \$180,000. What is its actual cost?

- A. \$198,000
- B. \$200,000
- C. \$180,000
- D. \$162,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A software rollout has a BAC of \$1,200,000. To date, EV is \$400,000 and AC is \$500,000, and the team expects current cost performance to continue. The vendor's sales rep has promised a discount next quarter, not yet in writing. What is the EAC?

- A. \$1,500,000
- B. \$960,000
- C. \$1,300,000
- D. \$1,200,000, because the promised discount will offset the overrun.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A project's BAC is \$600,000, and its EAC, assuming current performance continues, is \$750,000. What is its CPI?

- A. 0.20
- B. -\$150,000
- C. 1.25
- D. 0.80

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A marketing-system project has a BAC of \$800,000 and a current EAC of \$920,000. The steering committee meets monthly in the downtown office. What is the variance at completion?

- A. \$920,000
- B. -\$120,000, an expected overrun.
- C. +\$120,000, an expected savings.
- D. -\$120,000, meaning the project is behind schedule.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A project has a BAC of \$500,000, EV of \$200,000, and AC of \$250,000. Management insists it finish within the original BAC. The project manager just returned from a two-week leave. What CPI must the remaining work achieve?

- A. 1.20
- B. 0.83
- C. 1.00, because finishing on budget always requires exactly 1.0.
- D. 0.80

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A project needs a TCPI of 1.25 to finish within its BAC of \$400,000. Actual cost to date is \$240,000. What is earned value to date?

- A. \$200,000
- B. \$240,000
- C. \$272,000
- D. \$160,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A client asks a developer to add a reporting feature “since you're already in the code.” The developer estimates 30 hours and is known as the fastest coder on the team. The feature is not in the approved scope. What should happen?

- A.** Have the fastest coder build it after hours so the budget is not affected.
- B.** Add it now, since 30 hours is small.
- C.** Refuse every client request once the baseline is set.
- D.** Submit a change request and assess cost, schedule, and risk impact before any approval.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12.

A company has spent \$2 million on a custom CRM. Finishing it will take \$1.5 million more and deliver benefits worth \$1.2 million. An off-the-shelf tool would deliver the same benefits for \$0.4 million. The CIO championed the custom project at a board meeting. Based on cost, what is the best decision?

- A.** Finish the custom CRM, because \$2 million is already invested.
- B.** Do neither, since total spending of \$3.5 million exceeds the \$1.2 million benefit.
- C.** Switch to the off-the-shelf tool; the \$2 million is sunk and does not change the future comparison.
- D.** Finish the custom CRM to protect the CIO's standing with the board.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A city IT department is estimating a new fiber-network rollout. Reliable records show similar installs cost \$18 per linear foot, and the new network is 40,000 feet long. ~~The department head drives a hybrid and prefers vendor-neutral designs.~~ Which technique and estimate fit?

Target word: technique and estimate **Decides:** A unit rate times quantity is parametric.

A	Wrong stem number	\$18 is the unit rate, and no work packages were summed.
B	Inverted operation	Divides 40,000 by 18 instead of multiplying.
C	Correct	$\$18 \times 40,000 \text{ feet} = \$720,000$ using a statistical unit rate.
D	Confusion set	Analogous scales a whole past project's total; a per-unit rate is parametric.

2. Answer C Concept 2

A project's work-package estimates total \$900,000. The team adds \$60,000 of contingency for identified risks, and leadership holds \$40,000 of management reserve for unknown risks. ~~The sponsor is a former CFO who reviews every invoice.~~ What is the cost baseline?

Target word: cost baseline **Decides:** Baseline includes contingency but not management reserve.

A	Confusion set	Adds management reserve instead of contingency.
B	Dropped element	Leaves out the contingency reserve.
C	Correct	$\$900,000 + \$60,000 \text{ contingency} = \$960,000$.
D	Adjacent category	That is the total project budget, which adds management reserve.

3. Answer B Concept 2 · transfer

A project budget is \$2.5 million, including \$200,000 of management reserve. Contingency reserve inside the baseline is \$150,000. What is the cost baseline?

Target word: work backward from budget to baseline **Decides:** Baseline = budget – management reserve.

A	Wrong stem number	That is the full budget.
B	Correct	$\$2.5\text{M} - \$0.2\text{M management reserve} = \2.3M ; contingency stays in.
C	Adjacent category	Also removes contingency, which belongs in the baseline.
D	Inverted operation	Adds the management reserve instead of removing it.

4. Answer D Concept 3

At the halfway review, a warehouse-automation project has EV of \$300,000, AC of \$375,000, and PV of \$320,000. ~~The project team just won an internal innovation award.~~ What is the CPI, and what does it mean?

Target word: CPI and its meaning **Decides:** $\text{CPI} = \text{EV} / \text{AC}$.

A	Inverted operation	AC / EV flips the ratio.
B	Confusion set	EV / PV is SPI, a schedule measure.
C	Adjacent category	That is cost variance, not the index.
D	Correct	$\$300,000 / \$375,000 = 0.80$.

5. Answer B Concept 3 • transfer

A project's CPI is 0.90 and its earned value is \$180,000. What is its actual cost?

Target word: solve for AC **Decides:** $AC = EV / CPI$.

A	Confusion set	Treats dividing by 0.90 as adding 10%.
B	Correct	$\$180,000 / 0.90 = \$200,000$.
C	Wrong stem number	That is earned value.
D	Inverted operation	Multiplies EV by CPI instead of dividing.

6. Answer A Concept 4

A software rollout has a BAC of \$1,200,000. To date, EV is \$400,000 and AC is \$500,000, and the team expects current cost performance to continue. ~~The vendor's sales rep has promised a discount next quarter, not yet in writing.~~ What is the EAC?

Target word: EAC with typical variance **Decides:** $CPI = 0.80$; $EAC = BAC / CPI$.

A	Correct	$CPI = 400,000 / 500,000 = 0.80$; $\$1,200,000 / 0.80 = \$1,500,000$.
B	Inverted operation	Multiplies BAC by CPI instead of dividing.
C	Adjacent category	$AC + (BAC - EV)$ assumes the rest goes to plan, which the stem rules out.
D	Camouflage	An unwritten promise does not change the forecast.

7. Answer D Concept 4 • transfer

A project's BAC is \$600,000, and its EAC, assuming current performance continues, is \$750,000. What is its CPI?

Target word: solve for CPI **Decides:** $CPI = BAC / EAC$.

A	Dropped element	Overrun / EAC measures the overrun share, not the index.
B	Adjacent category	That is VAC, not CPI.
C	Inverted operation	EAC / BAC flips the ratio.
D	Correct	$\$600,000 / \$750,000 = 0.80$.

8. Answer B Concept 5

A marketing-system project has a BAC of \$800,000 and a current EAC of \$920,000. ~~The steering committee meets monthly in the downtown office.~~ What is the variance at completion?

Target word: VAC and its meaning **Decides:** $VAC = BAC - EAC$.

A	Wrong stem number	That is the EAC.
B	Correct	$\$800,000 - \$920,000 = -\$120,000$.
C	Inverted operation	Subtracts in the wrong order.
D	Confusion set	VAC is a cost measure, not a schedule measure.

9. Answer A Concept 6

A project has a BAC of \$500,000, EV of \$200,000, and AC of \$250,000. Management insists it finish within the original BAC. ~~The project manager just returned from a two-week leave.~~ What CPI must the remaining work achieve?

Target word: TCPI to BAC **Decides:** $TCPI = (BAC - EV) / (BAC - AC)$.

A	Correct	$(500,000 - 200,000) / (500,000 - 250,000) = 300,000 / 250,000 = 1.20$.
B	Inverted operation	Remaining funds / remaining work flips the ratio.
C	Absolute language	Once over budget, remaining work must beat 1.0 to recover.
D	Adjacent category	That is the current CPI, not what is needed going forward.

10. Answer A Concept 6 · transfer

A project needs a TCPI of 1.25 to finish within its BAC of \$400,000. Actual cost to date is \$240,000. What is earned value to date?

Target word: solve for EV from TCPI **Decides:** Remaining work = $1.25 \times$ remaining funds.

A	Correct	Remaining funds = \$160,000; remaining work = $1.25 \times \$160,000 = \$200,000$; EV = $\$400,000 - \$200,000$.
B	Wrong stem number	That is actual cost.
C	Inverted operation	Divides remaining funds by 1.25 instead of multiplying.
D	Confusion set	That is the remaining funds (BAC – AC), not EV.

11. Answer D Concept 7

A client asks a developer to add a reporting feature “since you’re already in the code.” The developer estimates 30 hours ~~and is known as the fastest coder on the team.~~ The feature is not in the approved scope. What should happen?

Target word: handling an out-of-scope request **Decides:** The feature is outside approved scope.

A	Camouflage	Speed does not remove the need for approval, and hidden hours are still cost.
B	Dropped element	Skips impact analysis and approval; this is scope creep.
C	Absolute language	Changes can be approved; they just need the process.
D	Correct	Integrated change control protects the baseline.

12. Answer C Concept 8

A company has spent \$2 million on a custom CRM. Finishing it will take \$1.5 million more and deliver benefits worth \$1.2 million. An off-the-shelf tool would deliver the same benefits for \$0.4 million. ~~The CIO championed the custom project at a board meeting.~~ Based on cost, what is the best decision?

Target word: go-forward decision **Decides:** The \$2 million is sunk; compare only future costs.

A	Confusion set	The sunk-cost fallacy.
B	Adjacent category	Includes sunk dollars in a forward-looking comparison.
C	Correct	Future cost \$0.4M vs. \$1.5M for the same \$1.2M benefit.
D	Camouflage	Reputation is not a cost factor in this comparison.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Cost estimating techniques	Analogous uses a similar past project's total (fast, rough); parametric multiplies a unit rate by quantity; bottom-up sums work-package estimates (slowest, most accurate).	Calling any use of historical data "analogous."	A per-unit rate and a quantity in the stem.
2	Cost baseline vs. project budget	Cost baseline = work estimates + contingency reserve; project budget = cost baseline + management reserve.	Putting management reserve inside the baseline.	Contingency and management reserve both given.
3	Cost variance and CPI	$CV = EV - AC$ and $CPI = EV / AC$; below 1 (or negative CV) means over budget.	Using PV in cost measures, or flipping the ratio.	EV and AC with a budget question.
4	Estimate at completion	If current cost performance continues, $EAC = BAC / CPI$.	Multiplying BAC by CPI, or assuming the rest will go to plan.	"Current performance expected to continue."
5	Variance at completion	$VAC = BAC - EAC$; negative means an expected overrun.	Reversing the sign or reading VAC as a schedule measure.	BAC and EAC given together.
6	To-complete performance index	$TCPI = (BAC - EV) / (BAC - AC)$: the CPI remaining work must achieve to finish on budget.	Assuming the current CPI or 1.0 is enough.	"Must finish within the original budget."
7	Change control for cost and scope	Any change to approved scope or cost goes through a change request with impact analysis before approval.	Adding "small" requests directly to the work.	A stakeholder asking a team member for extra work.
8	Sunk costs	Money already spent cannot be recovered and should not influence a go-forward decision; compare only future costs and benefits.	Continuing because so much has already been spent.	"Has already spent" next to a choice about the future.

Course scope (public description): The Budgeting and Cost Control in Project Management course equips learners to effectively manage project budgets and control costs throughout the project lifecycle. Learners create accurate project budgets using resource requirements, historical data, and cost estimation techniques, as well as explore techniques for monitoring project costs, analyzing cost variances, and implementing cost control measures to ensure projects stay within budget constraints. Learners will develop the skills necessary to manage project budgets, identify cost variances, and implement cost control measures to ensure project success, emphasizing the importance of change management, risk management, and proactive cost control strategies in achieving project objectives within budget constraints.